

ISHU SINGH

AI / Automation Engineer · InSAR & Geospatial · Full-Stack

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Portfolio: [ishunteam-png.github.io/portfolio-site](#) · GIS: [ishunteam-png.github.io/gis-portfolio](#)

PROFESSIONAL SUMMARY

AI/Automation engineer who builds production-grade software and geospatial pipelines. Delivered AI-assisted web apps, Telegram-driven workflow automations, healthcare platforms with 4 live hospital portals, and Sentinel-1 InSAR pipelines (2,655 persistent scatterers, EGMS-L3 equivalent road-subsidence analysis on a 3-year stack). Ships fast, works globally, available remotely.

5	2	4	7	17	2,655
Projects Shipped	Live Commercial	Hospital Portals	n8n Workflows	GIS Dashboards	InSAR PS

CORE SKILLS

Languages & Frameworks	Python, TypeScript, JavaScript, React, HTML/CSS
Geospatial / Remote Sensing	GeoPandas, rasterio, MintPy, SNAP, Google Earth Engine, Sentinel-1/2/5P, InSAR, OSMnx, NetworkX, Shapely, h3-py, pystac-client, xarray, LandTrendr, TROPOMI
Machine Learning / AI	scikit-learn, PyTorch, YOLOv8 (ultralytics), OpenCV
Data Engineering	DuckDB, pandas, pyarrow, scipy
Mapping / Visualization	Leaflet.js, Chroma.js, pvlb, PVGIS
Automation & Tooling	n8n, OR-Tools, HRRR, LANDFIRE, VIIRS
Web / DevOps	Vite, Tailwind CSS, GitHub Pages, GitHub Actions

PROFESSIONAL PROJECTS

Haimcore — AI Product [Shipped]

AI-assisted platform built with production-grade web app architecture. Implemented audit logs, rollback paths, and idempotency guards. Full-stack delivery from design to deployment.

Stack: Next.js · TypeScript · Anthropic SDK · React

Relief Guru — AI Workflow Automation [Shipped]

Telegram-driven n8n workflow automation system orchestrating 7 workflows in production. Integrated AI decision routing, HeyGen video generation, and Pinecone vector search.

Stack: n8n v4 · Telegram · HeyGen · Pinecone

Jalaram Healthcare Platform [4 Portals Live]

Full-stack healthcare web application with SystemLog auditing across 4 live hospital portals. Handles patient management, appointments, and staff workflows at scale.

Stack: React · TypeScript · Node.js · PostgreSQL

EDUCATION

Georgian Technical University — Tbilisi, Georgia

Bachelor of Science, Geodesy and Geoinformatics Engineering · Mar 2022 – Apr 2026

Freelance mapping and GIS portfolio development; built 17-project geospatial showcase deployed on GitHub Pages.

GIS / GEOSPATIAL PORTFOLIO — 17 Live Interactive Dashboards

[ishunteam-png.github.io/gis-portfolio](#) · Scan any QR code to open the live Leaflet dashboard

InSAR Road Subsidence Delhi, India INSAR 2,555 PS · V_U −10.3 to +4.7 mm/yr EGMS-L3 Sentinel-1 InSAR pilot with true 2D vertical + east-west decomposition. Velocity-threshold filter, temporal-coherence gate, live histogram. Sentinel-1 MintPy SNAP PyAPS3 https://ishunteam-png.github.io/gis-portfolio/01-insar-road-subsidence/	Café Location Intelligence Tbilisi, Georgia ROUTING 8,415 cells · 62,849-node graph · $\rho = 0.39$ Multi-signal suitability mapping over 504 km². 5 demand signals including walkability from a 62,849-node OSM pedestrian graph. OSMnx GeoPandas scipy Leaflet https://ishunteam-png.github.io/gis-portfolio/02-real-estate-intel/
Advanced VRP — Time Windows Benchmark dataset ROUTING 60 stops · 5 vehicles · OR-Tools −20.3% vs greedy Greedy vs Clarke-Wright vs OR-Tools GLS on 60 stops / 5 vehicles. OR-Tools wins by −20.3% on drive time. OSMnx NetworkX OR-Tools Leaflet https://ishunteam-png.github.io/gis-portfolio/03-route-optimization/	Jakarta Compound Flood Risk Jakarta, Indonesia HAZARD 1,360 schools · 138 hospitals · 4-factor index HAND + slope + drainage density + imperviousness risk index. 1,360 schools and 138 hospitals intersected. rasterio scipy.ndimage OSMnx GeoPandas https://ishunteam-png.github.io/gis-portfolio/04-flood-risk/
Web GIS Dashboard — Delhi PS FLAGSHIP Delhi, India INSAR ~480 lines vanilla JS · 6 presets · URL-hash state Flagship single-file Leaflet build. 6 colour presets, split-panel view, live velocity histogram, URL-hash state for shareable views. Leaflet Chroma.js Vanilla JS GeoJSON https://ishunteam-png.github.io/gis-portfolio/05-web-gis-dashboard/	GeoAI Road Damage × InSAR Delhi, India ML/GEOAI YOLOv8m-seg · all 4 detections on subsiding PS YOLOv8m-seg correlated with subsiding persistent scatterers. Operationalises the InSAR → surface damage causal chain. ultralytics PyTorch OpenCV Leaflet https://ishunteam-png.github.io/gis-portfolio/06-geoai-road-damage/
Sentinel-2 LULC — Bengaluru Bengaluru, India ML/GEOAI OA 90.9% · κ 0.89 · built-up +22.6% (2020→2024) Random Forest on 6 bands + 7 spectral indices. Built-up grew +22.6%; bare soil dropped 24% across 4 years. Sentinel-2 pystac-client scikit-learn rasterio https://ishunteam-png.github.io/gis-portfolio/07-lulc-classification/	15-Minute City Paris, France ROUTING 87.7% cells ≥5/6 categories · 47,812-node graph Carlos Moreno's 6-category framework over Paris. 47,812-node OSM walk graph; per-cell Dijkstra at 1,250 m. OSMnx NetworkX Shapely alphashape https://ishunteam-png.github.io/gis-portfolio/08-15min-city/
NDVI Deforestation — Rondônia Rondônia, Brazil ENVIRONMENT 2,698 km² cleared 2015–2024 · 2019 fire spike visible Per-pixel breakpoint detection over the BR-364 fishbone corridor. 2019 fire spike and 2024 policy drop visible in the loss curve. Earth Engine Landsat 8/9 Sentinel-2 LandTrendr https://ishunteam-png.github.io/gis-portfolio/09-ndvi-change/	H3 Mobility — NYC Yellow Taxi New York City, USA MOBILITY 57,000 daily trips · H3 res-9 · 5 metrics × 5 windows TLC pickups/dropoffs indexed to H3 res-9 hexagons. AM commute-in / PM airport-dropoff asymmetry surfaces across 5 time windows. h3-py DuckDB pandas pyarrow https://ishunteam-png.github.io/gis-portfolio/10-h3-mobility/
Wildfire Spread — Park Fire CA California, USA HAZARD 961/1,024 cells burned by h120 · peak 28 m/h Rothermel cellular automaton on a 32×32 LANDFIRE fuel grid with HRRR NE Diablo wind forcing. Hourly slider, wind-vector overlay. Rothermel CA LANDFIRE HRRR wind VIIRS https://ishunteam-png.github.io/gis-portfolio/11-wildfire-spread/	Rooftop Solar Potential Lisbon, Portugal ENVIRONMENT 12.5 GWh/yr · 450 rooftops · ≐ 4,178 households OSM polygons × Copernicus DEM × pvlib irradiance × skyline-shading raycast over 450 rooftops in 5 neighbourhoods. OSMnx py3dep pvlib PVGIS https://ishunteam-png.github.io/gis-portfolio/12-solar-lisbon/

Sentinel-5P NO₂ Plume

Delhi, India

ENVIRONMENT

100% WHO exceedance Jan · 10% Jun · 12-month slider

TROPOMI L3 monthly composites; 728 cells, 11 CPCB hotspots, Gaussian plume. Winter inversion vs monsoon scrubbing visible.

Sentinel-5P | TROPOMI | xarray | CPCB

<https://ishunteam-png.github.io/gis-portfolio/13-no2-delhi/>

AIS Vessel Tracking

Singapore Strait

MOBILITY

978 vessels · 200 anchored · +18% Red Sea backlog

H3 res-8 density grid + DBSCAN anchorage clusters. 2024 Red Sea diversion drove +18% congestion backlog Jan–Apr.

AIS | H3 res-8 | DBSCAN | MarineTraffic

<https://ishunteam-png.github.io/gis-portfolio/14-ais-singapore/>

S1+S2 Crop Classification

Story County, Iowa, USA

ML/GEOAI

S2-only 86.3% → S1+S2 fusion 92.7% (+6.4% uplift)

Sentinel-1 SAR (VV/VH) + Sentinel-2 NDVI composites over 672 fields, 7 CDL classes. Toggle between truth, S1-only, S2-only, fused.

Sentinel-1 | Sentinel-2 | scikit-learn | USDA CDL

<https://ishunteam-png.github.io/gis-portfolio/15-crop-classification/>

Urban Heat Island

Tokyo, Japan

ENVIRONMENT

UHI 5.6°C · peak 37.4°C · $\rho = -0.82$ with NDVI

Landsat-9 thermal LST × Sentinel-2 NDVI over the August 2024 heat-wave peak. More trees = cooler block, quantified at city scale.

Landsat 9 | Sentinel-2 | pystac-client | scipy.stats

<https://ishunteam-png.github.io/gis-portfolio/16-uhi-tokyo/>

Coastal Erosion — Florida

Florida Atlantic, USA

ML/GEOAI

Daytona −27.4 m · Miami Beach +24.0 m · 2015–2024

Annual NDWI shoreline migration across 15 stations, 2015–2024. Hurricane Irma 2017 visible. Miami Beach +24.0 m from renourishment.

Sentinel-2 NDWI | scikit-image | Shapely | FDEP transects

<https://ishunteam-png.github.io/gis-portfolio/17-coastal-erosion-florida/>

All 17 dashboards are live on GitHub Pages · Scan QR codes to open directly · Full source at github.com/ishunteam-png/gis-portfolio